

Literature Status: Positivity of the Real ℓ_p Numerical Index

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2026-06-14

Status

This is a literature-already-answered packet, not an original proof from the run. Problem 1 in Kadets–Martin–Paya, arXiv:math/0605781, asks whether the numerical index $n(\ell_p)$ is positive for every real ℓ_p with $p \neq 2$. Martin–Meri–Popov, arXiv:0903.2704, answer this affirmatively in the stronger form

$$n(L_p(\mu)) > 0 \quad (1 < p < \infty, p \neq 2)$$

for every positive measure μ in the real case.

Original Problem

The original source is:

V. Kadets, M. Martin, R. Paya, *Recent progress and open questions on the numerical index of Banach spaces*, arXiv:math/0605781.

After recalling that

$$n(L_p[0, 1]) = n(\ell_p) = \inf_m n(\ell_p^m),$$

the paper states that $n(\ell_p^m) > 0$ for each finite m and each $p \neq 2$, but that the infinite-dimensional positivity was unknown:

THEOREM 1 ([22, 22]). *Let $1 \leq p \leq \infty$ be fixed. Then,*

- (a) $n(L_p[0, 1]) = n(\ell_p) = \inf\{n(\ell_p^m) : m \in \mathbb{N}\}$, and the sequence $\{n(\ell_p^m)\}_{m \in \mathbb{N}}$ is decreasing.
- (b) $n(L_p(\mu)) \geq n(\ell_p)$ for every positive measure μ .
- (c) In the real case,

$$\frac{1}{2}M_p \leq n(\ell_p^2) \leq M_p, \quad \text{where } M_p = \sup_{t \in [0, 1]} \frac{|t^{p-1} - t|}{1 + t^p}.$$

When $p \neq 2$, it is known that $n(\ell_p^m) > 0$ for every $m \geq 2$, and also that $v(T) > 0$ for every non-null $T \in L(\ell_p)$ (see [22, Theorem 2.3 and subsequent remark]), but we do not know if $n(\ell_p) > 0$. Observe that a positive answer to this question implies, thanks to (b) above, that $n(L_p(\mu)) > 0$ for every positive measure μ .

Problem 1. *Is $n(\ell_p)$ positive for every $p \neq 2$?*

With respect to item (c) in the above theorem, let us explain the meaning of the number M_p . It is deduced from [20, §2] that, since an operator $T \in L(\ell_2)$ is dominated by the operator $(a - b)$

Later Answer

The supporting source is:

M. Martin, J. Meri, M. Popov, *On the numerical index of real $L_p(\mu)$ -spaces*, arXiv:0903.2704.

Corollary 3 gives the quantitative lower bound

$$n(L_p(\mu)) \geq \frac{M_p}{12e}, \quad M_p = \max_t \frac{|t^{p-1} - t|}{1 + t^p}.$$

Since $M_p > 0$ when $p \neq 2$, Corollary 4 states the positive answer for all real $L_p(\mu)$ spaces, $1 < p < \infty$, $p \neq 2$, and explicitly says that this answers the question also posed in reference [7, Problem 1]. Reference [7] is the 2006 Kadets–Martin–Paya survey.

$$2|v|(T) \geq v(T_{\mathbb{C}}) \geq n(L_p^{\mathbb{C}}(\mu))\|T_{\mathbb{C}}\| \geq n(L_p^{\mathbb{C}}(\mu))\|T\|. \quad \square$$

It remains to notice that $n(L_p^{\mathbb{C}}(\mu)) \geq 1/e$ (as happens for any complex Banach space, see [1, Theorem 4.1]), to get the following consequence from the above two theorems.

Corollary 3. *Let $1 < p < \infty$ and let μ be a positive measure. Then, in the real case, one has*

$$n(L_p(\mu)) \geq \frac{M_p}{12e}$$

$$\text{where } M_p = \max_{t \geq 1} \frac{|t^{p-1} - t|}{1 + t^p}.$$

Since, clearly, $M_p > 0$ for $p \neq 2$, we get the following consequence which answers in the positive a question raised by C. Finet and D. Li (see [5, 6]) also posed in [7, Problem 1].

Corollary 4. *Let $1 < p < \infty$, $p \neq 2$ and let μ be a positive measure. Then $n(L_p(\mu)) > 0$ in the real case. In other words, the numerical radius and the operator norm are equivalent on $\mathcal{L}(L_p(\mu))$.*

It is a particular case of [6, Theorem 2.2] that $n(L_p(\mu)) = \inf_m n(\ell_p^m)$ for every infinite-dimensional $L_p(\mu)$ -space. For finite-dimensional spaces, $n(\ell_p^m) \leq n(\ell_p^2)$ for every $m \geq 2$ since ℓ_p^2 is an ℓ_p -summand

Proof Intuition

The supporting result does more than the original ℓ_p question asks: it proves a uniform numerical-radius lower bound for every operator on every real $L_p(\mu)$. The constant is positive exactly outside the Hilbertian case $p = 2$. Taking μ to be counting measure on $\mathbb{N}$ specializes the theorem to ℓ_p , so the original problem follows immediately.

Verification Notes

The answer is not contained in the original 2006 paper; there the positivity of $n(\ell_p)$ is posed as Problem 1. The supporting paper is a separate later paper and explicitly cites the 2006 survey as [7]. Its Corollary 4 answers [7, Problem 1] in the positive.

The endpoint cases $p = 1$ and $p = \infty$ were already standard in the 2006 source, because L -spaces and M -spaces have numerical index 1. Thus the supporting theorem resolves the only nontrivial range of the question.

Search Bounds

The local registry, solution index, and attempt index were checked for arXiv:0605781, arXiv:0903.2704, positivity of $n(\ell_p)$, and the adjacent two-dimensional formula. A web search for numerical-index positivity of ℓ_p and L_p spaces located arXiv:0903.2704. The arXiv page, source paper, and supporting PDF were inspected.

Limitations

This packet does not claim an original proof. It also does not address the neighboring 2006 questions asking whether $n(\ell_p) = n(\ell_p^2)$ or asking for the complex two-dimensional formula.

Human Review Recommendation

Check two points: first, that [7] in arXiv:0903.2704 is indeed the Kadets–Martin–Paya survey; second, that the specialization of real $L_p(\mu)$ to ℓ_p via counting measure matches the convention in the original problem.